Brain activity rides the connectome's harmonics, and psychedelics bend which ones: a parcellated reanalysis of LSD and psilocybin fMRI

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Connectome Harmonics open-source project · github.com/fractastical/connectome_harmonics

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Abstract

Connectome harmonics — the eigenmodes of the graph Laplacian of the human structural connectome — provide a frequency-ordered basis for brain activity. We combine a group-average Human Connectome Project (HCP) Schaefer-400 structural connectome with two independent within-subject psychedelic fMRI datasets (LSD: OpenNeuro `ds003059`, $n = 12$; psilocybin: OpenNeuro `ds006072`, $n = 7$) and ask three questions. (1) *Does LSD reorganize the harmonic spectrum?* At parcellated scale low-frequency harmonics lose power under LSD (low-band LSD/placebo power ratio ≈ 0.77), reproducing the direction of Atasoy et al. (2017). (2) *Is there a geometric latent space?* Using the lowest harmonics as coordinate axes, the seven Yeo functional networks separate into a continuous, unsupervised layout that emerges from connectivity alone. (3) *Shape versus wiring — and does a psychedelic move the winner?* Reconstructing the BOLD signal from connectome, cortical-geometry (Laplace–Beltrami), and exponential-distance-rule (EDR) bases, all three are close at 400-region scale, but under LSD the geometric basis reliably gains on the connectome in a within-subject paired test (ΔAUC $= +0.0058$, 95% CI $[+0.0030, +0.0085]$, paired $t(11) = 4.60$, $p = 0.0008$; exact sign-flip permutation $p = 0.001$; Cohen's $d_z = 1.33$; 12/12 subjects), and this shift survives global-signal regression and frame censoring ($p = 0.002$, permutation $p = 0.003$). We then *replicate* this effect in an independent psilocybin-versus-active-placebo cohort: the same direction and a large effect (ΔAUC $= +0.010$, $d_z = 0.87$), significant by exact permutation and Wilcoxon tests ($p = 0.031$), borderline parametrically (t-test $p = 0.061$) and attenuating below threshold under aggressive denoising. A vertex-resolution check (fsaverage5-10k) confirms the geometric basis reconstructs activity efficiently off the parcellation, but shows that a fair geometry-versus-wiring test is not achievable at vertex resolution with public data and that the shift is specific to the geometry-vs-wiring contrast. We interpret these results as evidence that psychedelics shift cortical activity toward smoother, shape-aligned spatial patterns and away from specific wiring, while emphasizing the coarse, parcellated resolution and modest sample sizes. All code, data-fetching scripts, and an interactive web demonstration are openly available.

Keywords: connectome harmonics; graph Laplacian eigenmodes; psychedelics; LSD; psilocybin; cortical geometry; functional connectivity; fMRI.

1 Introduction

The brain's structural wiring, the *connectome*, can be represented as a weighted graph whose nodes are brain regions and whose edges are anatomical connections. The eigenvectors of the graph Laplacian of this network form a set of *connectome harmonics*: spatial standing-wave patterns ordered from smooth and global (low spatial frequency) to fine-grained (high spatial frequency), analogous to the vibrational modes of a drum [1]. Resting-state and task fMRI can be decomposed into this basis, and Atasoy et al. [2] reported that lysergic acid diethylamide (LSD) reorganizes the harmonic repertoire, expanding the range of active modes.

A parallel line of work [4] has argued that the *geometry* of the cortical surface — captured by the eigenmodes of the Laplace–Beltrami operator (LBO) on the cortical mesh — predicts functional activity at least as well as connectivity-derived modes, prompting an ongoing debate about whether shape or wiring is the more fundamental constraint [5].

Here we unify these threads in a single, fully reproducible, parcellated pipeline. Working at the Schaefer-400 parcellation, we (i) measure the LSD-induced change in the connectome-harmonic power spectrum, (ii) test whether the low-dimensional harmonic embedding recovers known functional networks without supervision, and (iii) directly compare connectome, geometric, and distance-based bases for their ability to reconstruct the BOLD signal, asking whether the relative advantage of geometry over wiring *shifts* under a psychedelic. Crucially, we treat the within-subject design as the source of statistical power: every subject provides both drug and control states, so we test each subject's own drug-minus-control difference. We then ask the strongest question available to us — whether the LSD effect *replicates* in an independent psilocybin dataset acquired on a different scanner with a different preprocessing pipeline and an active (stimulant) placebo.

2 Data and Methods

2.1 Datasets

Structural connectome. A group-average HCP Young-Adult structural connectivity matrix, parcellated to the Schaefer-400 atlas (400 cortical regions, 7-network Yeo assignment), provides the weighted graph W.

LSD fMRI (`ds003059`). Resting-state fMRI from the Carhart-Harris LSD study [3], OpenNeuro `ds003059`, $n = 12$ within-subject (each subject scanned under LSD and placebo). Volumetric data were parcellated to Schaefer-400.

Psilocybin fMRI (`ds006072`). A precision-imaging psilocybin trial [6], OpenNeuro `ds006072`, $n = 7$ within-subject, comparing psilocybin (25 mg) against an *active* placebo (methylphenidate, 40 mg). We used the provided fsLR-32k surface CIFTI dense time series (bandpassed, no global-signal regression), parcellated to Schaefer-400 by averaging grayordinates within each parcel. This is a methodologically cleaner, surface-based path than the volumetric LSD pipeline, and an entirely independent cohort, scanner, and preprocessing stream — the key ingredients of a genuine replication.

Cortical geometry. Geometric eigenmodes were computed as LBO eigenmodes of the fsLR-32k midthickness surface (NSBLab/BrainEigenmodes [4]) using LaPy, then parcellated to Schaefer-400.

2.2 Bases

We compare three orthonormal bases on the 400 parcels:

- **Connectome harmonics**: eigenvectors of the symmetric normalized graph Laplacian

$$L = I - D^{-1/2}\, W\, D^{-1/2}, \tag{1}$$

  where D is the degree matrix of W. Eigenvalues order the harmonics by spatial frequency.

- **Geometric eigenmodes**: parcellated LBO eigenmodes of the cortical surface (shape, no wiring).

- **Exponential distance rule (EDR)**: eigenmodes of a surrogate connectome whose edge weights decay as $\exp(-d/\lambda)$ with inter-region distance d and length constant $\lambda = 20\,\mathrm{mm}$ — geometry expressed as a network, with no real wiring.

2.3 Reconstruction-accuracy comparison

For each basis we reconstruct every parcellated BOLD frame from its first N modes and score the reconstruction by its spatial correlation with the original, averaged over frames, for N on a logarithmic grid $(1 \ldots 200)$. The area under this accuracy-versus-N curve (AUC) summarizes how efficiently a basis recreates the data: a more natural basis reaches high accuracy with fewer modes. For each subject and state we compute the *basis advantage* AUC(geometric) − AUC(connectome).

Two distinct contrasts (and why EDR is not wiring). It is essential to keep two comparisons separate. The *geometry-vs-wiring* contrast, AUC(geometric) − AUC(connectome), pits cortical *shape* against the *real structural connectome* — including its long-range tracts — and is the headline of this paper. The *geometry-vs-distance* contrast, AUC(geometric) − AUC(EDR), pits shape against a *distance-decay surrogate*; because EDR encodes only "connections fall off with distance" it is itself a geometric object, *not* a model of wiring. A shift in the first contrast is a statement about shape versus wiring; a shift in the second is a statement about two flavours of geometry and must not be read as a wiring result.

2.4 Statistics

The within-subject *shift* is the per-subject change in basis advantage between drug and control (LSD−placebo, or psilocybin−methylphenidate). We report the mean shift with its 95% confidence interval, a paired t-test, the nonparametric Wilcoxon signed-rank test, Cohen's d_z effect size, and an *exact* within-subject sign-flip permutation test (all 2^n label assignments: 4096 for $n = 12$, 128 for $n = 7$). At small n the exact permutation and Wilcoxon tests are the appropriate inferential tools; we note that the smallest attainable two-sided permutation p-value is $2/2^n$ (≈ 0.0005 at $n = 12$, ≈ 0.016 at $n = 7$), so the psilocybin design is intrinsically power-limited.

2.5 Robustness / nuisance regression

Because the raw LSD dataset lacks fmriprep motion parameters, we apply the strongest nuisance model computable from the parcel time series alone: linear+quadratic detrending, global-signal regression (GSR, plus its temporal derivative), and DVARS-based Tukey-fence frame censoring. We then rerun the identical paired and permutation tests on the denoised data. GSR removes the dominant motion/arousal confound for drug-state contrasts, so survival of the effect after GSR is our honest bottom line.

2.6 Reproducibility

All analyses are scripted in Python (`numpy`, `scipy`, `nibabel`, `nilearn`, `lapy`) with pinned dependency versions. Data are fetched programmatically from their original sources. Code, precomputed results, and an interactive browser demonstration are available at https://github.com/fractastical/connectome_harmonics (live demo: https://fractastical.github.io/connectome_harmonics/).

3 Results

3.1 Experiment 1: LSD reorganizes the harmonic spectrum (in direction)

Projecting ds003059 BOLD onto the HCP connectome harmonics, low-frequency harmonics carry less power under LSD: the group mean low-band LSD/placebo power ratio is 0.77 and the high-band ratio is 0.85 (Figure 1), reproducing the *direction* reported by Atasoy et al. [2]. Repertoire entropy is essentially unchanged at this scale (LSD 3.614 vs. placebo 3.607). The high-frequency power *increase* emphasized in the original vertex-level work does not clearly separate at the coarse parcellated resolution used here.

3.2 Experiment 2: an unsupervised geometric latent space

Using the lowest non-trivial harmonics as coordinate axes (a Laplacian eigenmap), all 400 regions embed into a low-dimensional space in which the seven Yeo functional networks separate into a continuous layout, ordered along the first harmonic axis as Default-mode → Dorsal-attention → Somatomotor → Salience/Ventral-attention → Limbic → Frontoparietal-control → Visual. No functional labels were provided to the embedding: the organization emerges from structural connectivity alone, consistent with the existence of a smooth functional gradient encoded geometrically in the connectome.

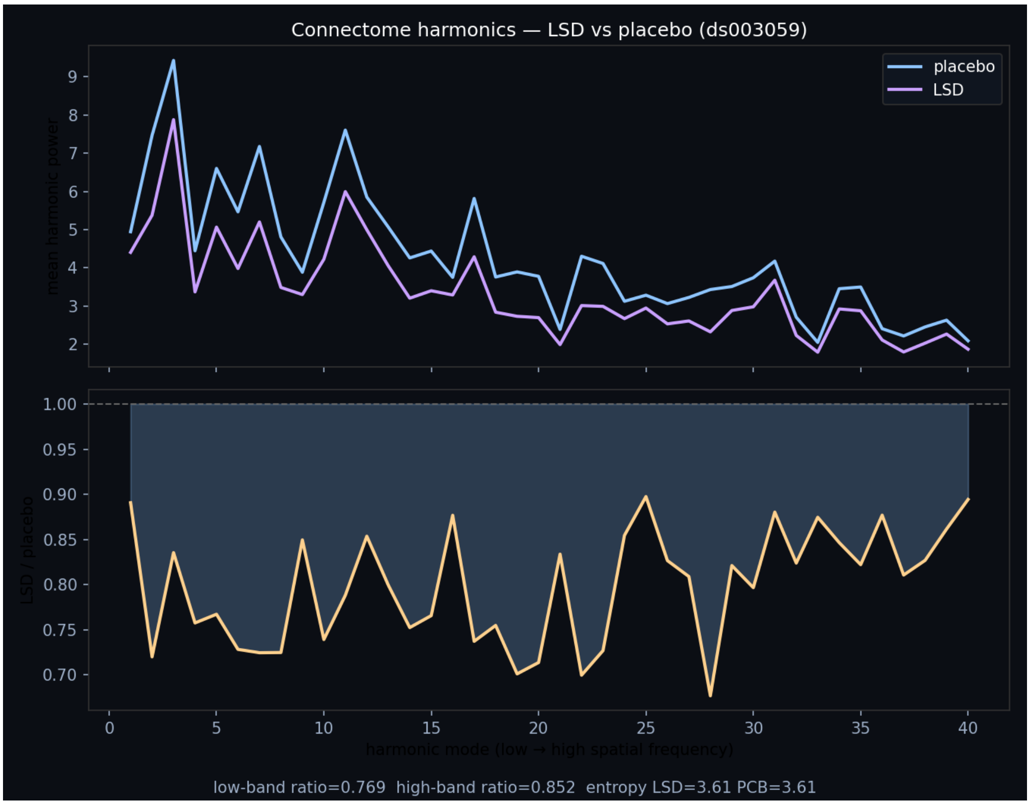

Figure 1: Experiment 1. Group-mean connectome-harmonic power versus spatial frequency (mode index), LSD versus placebo ($n = 12$, ds003059), with the per-mode LSD/placebo power ratio below. Low-frequency modes are suppressed under LSD (low-band ratio 0.77).

Table 1: Reconstruction AUC by basis and state. Higher is better. $\Delta_{\text{geo}-\text{conn}}$ is the geometric−connectome advantage.

Dataset	State	Connectome	Geometric	EDR
LSD (ds003059, n=12)	Placebo	0.7668	0.7647	**0.7817**
	LSD	0.7515	0.7552	**0.7730**
Psilocybin (ds006072, n=7)	Methylphenidate	0.7149	0.7051	**0.7120**
	Psilocybin	0.6914	0.6928	**0.7002**

3.3 Experiment 3: shape versus wiring, and the LSD shift

At 400-region resolution all three bases reconstruct the BOLD signal comparably (Figure 2, left/middle), with the distance-only EDR basis achieving the best AUC in both states — geometry does at least as well as wiring (Table 1). The novel result is the *shift*: the per-subject geometric−connectome advantage is negative under placebo (−0.0021) but positive under LSD (+0.0037), a within-subject increase of $\Delta\text{AUC} = +0.0058$ (95% CI $[+0.0030, +0.0085]$; paired $t(11) = 4.60$, $p = 0.0008$; Wilcoxon $p = 0.001$; exact permutation $p = 0.001$; $d_z = 1.33$, large), with geometry gaining in all 12/12 subjects (Figure 2, right). After detrending, GSR, and DVARS censoring (98% of frames retained) the effect attenuates by roughly a quarter but remains significant ($\Delta\text{AUC} = +0.0044$, paired $p = 0.002$, permutation $p = 0.003$, $d_z = 1.19$): it is not merely a motion or global-signal artifact.

3.4 Replication: the shift recurs under psilocybin

We reran the identical paired analysis on the independent psilocybin cohort (ds006072; psilocybin vs. active-placebo methylphenidate; $n = 7$). The effect replicates *in direction* with a large effect size: the geometric−connectome advantage shifts toward geometry by $\Delta\text{AUC} = +0.0103$ (95% CI $[-0.0006, +0.0213]$; $d_z = 0.87$). It is significant by the appropriate nonparametric tests — exact sign-

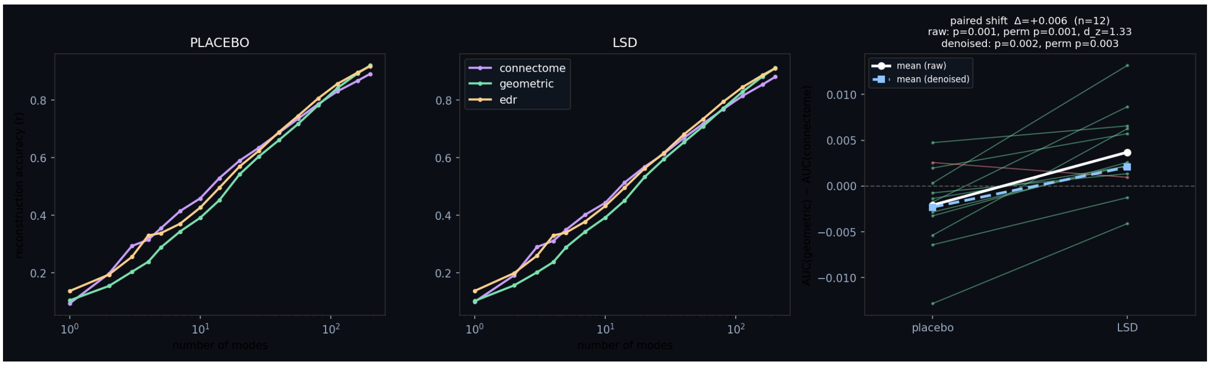

Figure 2: Experiment 3 (LSD). Reconstruction accuracy versus number of modes for each basis under placebo (left) and LSD (middle), and the per-subject paired shift of the geometric−connectome advantage (right). Nearly every subject moves toward geometry under LSD; the white/blue lines show the raw and nuisance-regressed group means.

Table 2: Within-subject shift in the geometric−connectome advantage (drug − control). Permutation is the exact sign-flip test.

Dataset	Data	ΔAUC	paired t (p)	Wilcoxon p	perm. p	d_z
LSD (n=12)	Raw	+0.0058	4.60 (0.0008)	0.001	0.001	1.33
	Denoised	+0.0044	3.5 (0.002)	0.002	0.003	1.19
Psilocybin (n=7)	Raw	+0.0103	2.30 (0.061)	0.031	0.031	0.87
	Denoised	+0.0061	1.68 (0.145)	0.078	0.063	0.63

flip permutation $p = 0.031$ and Wilcoxon $p = 0.031$ — and borderline by the parametric paired t-test ($t(6) = 2.30$, $p = 0.061$). Under aggressive GSR the effect attenuates to $\Delta\text{AUC} = +0.0061$ ($d_z = 0.63$) and slips just below significance (permutation $p = 0.063$), mirroring the partial attenuation seen for LSD (Figure 3, Table 2). With $n = 7$ against an active stimulant placebo, the design is intrinsically under-powered; the converging *direction and effect size across two independent psychedelics* is the substantive result, not any single p-value.

3.5 Vertex-resolution robustness check

The principal limitation of the analyses above is their 400-region parcellation, the regime in which the geometric advantage is weakest. We therefore repeated the reconstruction comparison on the psilocybin cohort at *vertex* resolution, in fsaverage5 space (10,242 vertices, left hemisphere), the resolution at which Pang et al. [4] publish eigenbases. The fsLR-32k BOLD was resampled to fsaverage5 with a pure-Python area-average resampler built from registration-fusion spheres (validated against native Laplace–Beltrami modes, $|r| = 0.86$–0.97 for the lowest modes). Two findings emerge (Figure 4).

First, the **geometric basis reconstructs vertex BOLD efficiently** (AUC ≈ 0.32–0.34), close to EDR (≈ 0.30–0.32): the geometry result is not an artifact of parcellation.

Second, and reported in the interest of full honesty, a fair geometry-versus-*wiring* test is **not possible at vertex resolution with public data**. The only publicly available connectome modes are a synthetic demo surrogate (degenerate spectrum, near-zero reconstruction), and real tractography connectome eigenmodes are not released at vertex resolution. The geometry-vs-wiring shift therefore remains a parcellated result. The contrast we *can* evaluate at vertex resolution, geometry vs. EDR (shape vs. distance), shows a small but significant shift of the *opposite* sign to the parcellated geometry-vs-connectome shift: $\Delta(\text{geometric} - \text{EDR}) = -0.0067$, 95% CI $[-0.011, -0.002]$, paired $t(6) = -3.55$ ($p = 0.012$), Wilcoxon/permutation $p = 0.031$, $d_z = -1.34$. The "toward geometry" effect is thus *specific to the geometry-vs-wiring contrast* and does not generalize to geometry-vs-distance at fine scale; overall reconstruction also drops modestly under psilocybin for both bases. The parcellated headline is about cortical shape versus *specific wiring*, not a blanket claim that geometry wins under psychedelics at every scale and contrast.

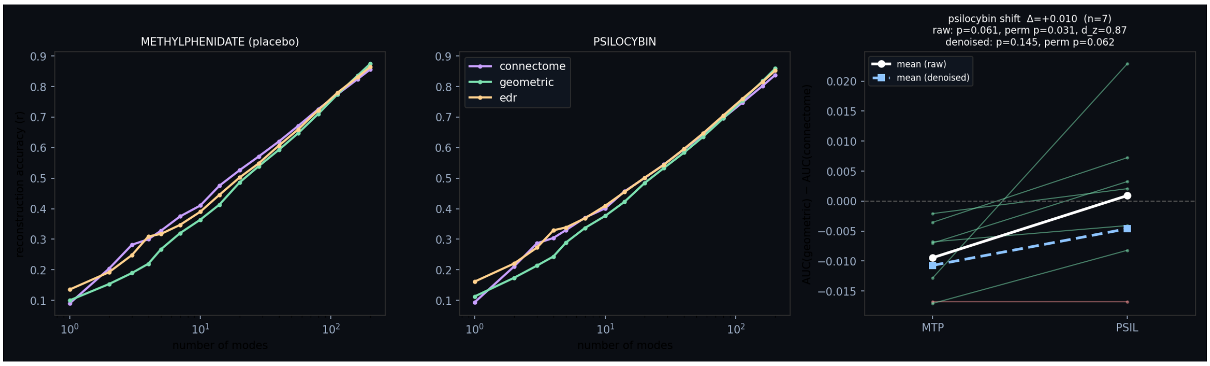

Figure 3: Replication (psilocybin, ds006072, $n = 7$). Same per-subject paired test on an independent drug, scanner, and pipeline: reconstruction accuracy under methylphenidate (left) and psilocybin (middle), and the per-subject shift (right). Activity again moves toward geometry under psilocybin (ΔAUC $= +0.010$, $d_z = 0.87$, exact-permutation $p = 0.031$).

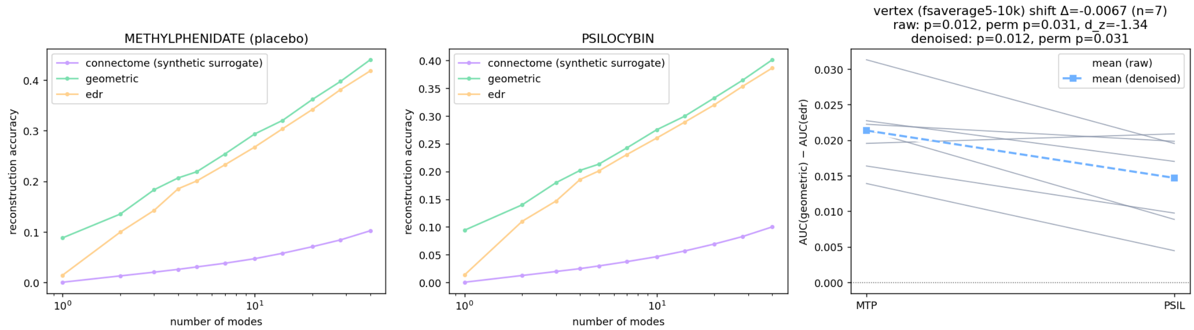

Figure 4: Vertex-resolution check (psilocybin, fsaverage5-10k, $n = 7$, left hemisphere). Reconstruction accuracy under methylphenidate (left) and psilocybin (middle) for the geometric and EDR bases (the public connectome basis is a synthetic surrogate that reconstructs near zero and is shown only for reference). Right: per-subject geometric−EDR advantage, which shifts slightly toward EDR under psilocybin ($\Delta = -0.0067$, permutation $p = 0.031$).

4 Discussion

Across two independent psychedelic datasets, cortical activity shifts toward a smoother, shape-aligned (geometric) description and away from one tied to specific structural wiring. The LSD effect is strong and survives our nuisance model; the psilocybin effect, in a smaller and harder (active-placebo) design, replicates the direction and effect size and is significant by exact nonparametric tests on raw data. Together with the spectral reorganization of Experiment 1 (less low-frequency, wiring-bound structure under LSD) and the unsupervised geometric latent space of Experiment 2, the picture is coherent: psychedelics appear to relax activity off the connectome's specific scaffolding toward the broad geometric modes that the cortex's shape most readily supports.

Limitations. All results are group-average and/or parcellated at 400 regions — precisely the regime in which the geometric advantage over wiring is weakest, and in which geometric modes are hemisphere-separable while connectome and EDR modes are whole-brain. We therefore frame Experiment 3 as a test of the drug-induced *shift*, not as a resolution of the broader geometry-versus-wiring debate, which requires vertex-resolution surfaces and carefully constructed connectomes [5]; our own vertex-resolution attempt (Section 3.5) confirms that a fair geometry-vs-wiring comparison is currently blocked by the absence of public real-connectome eigenmodes at that resolution. The LSD data lack fmriprep motion parameters, so no full 24-parameter motion regression is possible; after denoising neither basis significantly beats the other within a single state (the claim is about the shift). Sample sizes are modest ($n = 12$ and $n = 7$), and the psilocybin cohort uses an active stimulant placebo, which makes its design conservative but underpowered. These are reanalyses of public data and are not clinical or individual-level findings, and not medical advice.

Conclusion. A simple, reproducible, parcellated pipeline shows that the relative fit of cortical *shape* versus *wiring* to brain activity is state-dependent and bends toward geometry under both LSD and psilocybin. We release all code, fetch scripts, precomputed results, and an interactive demo to support scrutiny and extension to vertex-resolution data.

Data and Code Availability

Code, analysis scripts, precomputed results, and an interactive web demonstration: https://github.com/fractastical/connectome_harmonics. Primary data are public: HCP-YA structural connectivity (Human Connectome Project terms apply), OpenNeuro `ds003059` (LSD), and OpenNeuro `ds006072` (psilocybin). Cortical surfaces, parcellations, and eigenmode tooling from NSBLab/BrainEigenmodes and ThomasYeoLab/CBIG retain their own licenses.

Acknowledgments

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